Mohammed Abdul Momin Siddiqui

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.12
Narayana Junior College, Dilsukhnagar, Hyderabad	Class XII (Telangana State Board of Intermediate Education)	2021	98.40%
Maria Pia's Good Shepherd High School	Class X (Telangana State Board of Secondary Education)	2019	9.80%

PROJECTS

SWAT: Security-Oriented Warm-Up Tuning for LLMs | *GitHub*

- Reproduced the ACL 2026 SWAT alignment defense on LLaMA-2-7B, mitigating Large Language Model (LLM) safety degradation during Instruction Fine-Tuning (IFT).
- Mapped security-critical attention modules (Mods_Rob covering 28.12% of parameter slots) via multi-layer perceptron probing and perturbation scans.
- Orchestrated a two-phase LoRA pipeline using differential learning rates, **cutting Attack Success Rate to 16.82%** while improving **task accuracy by 19.00%**

Cross-Cultural Norm & Behavior Classifier (NLP course project) | *GitHub*

- Fine-tuned a 110M-parameter BERT transformer across ~44k balanced samples, achieving a **0.998 F1-score** classifying behavioral norms versus factual statements.
- Structured an **adversarial culture-masking pipeline** replacing demographic tokens with [CULTURE] tags to enforce semantic reasoning and eliminate dataset pattern leakage.
- Boosted 10-class attribution Macro F1 **from 0.60 to 0.83** (and accuracy from 72% to 89%) over an n-gram TF-IDF baseline, significantly increasing recall on low-resource classes (German F1: 0.33 to 0.79; Italian F1: 0.50 to 0.70).

Robust Compositional Visual Reasoning (IIT-H DL course hackathon) | *GitHub*

- Engineered an OpenCV contour preprocessing pipeline to eliminate metallic reflection shifts, lifting baseline classification accuracy from **~74% to over 90%**.
- Constructed a 21.4M-parameter SE-PreActResNet-34 integrating Squeeze-and-Excitation attention and stochastic depth, reaching **99.72% validation accuracy** via PyTorch mixed-precision training.
- Deployed a 4-way Test-Time Augmentation (TTA) ensemble across 5,010 unlabelled test samples to maximize boundary inference confidence.

Shredded Image Recovery Engine | *GitHub*

- Modeled pairwise seam compatibility using a Keras **Convolutional Neural Network (CNN)** to predict patch adjacency based on texture, text alignment, and edge gradients.
- Designed a **greedy graph-reassembly algorithm** that pruned factorial (N!) combinatorial search spaces to reconstruct shredded images.

TransactiWar - Secure Peer-to-Peer Financial Web Application | *GitHub*

- Architected a zero-framework, ACID-compliant ledger platform in vanilla PHP and PostgreSQL using row-level locking (SELECT ... FOR UPDATE) , preventing double-spend race conditions.
- Secured authentication endpoints using BCrypt hashing (cost factor 12), constant-time dummy hashing against timing attacks, and automated brute-force IP lockout policies.
- Built a zero-trust media pipeline validating MIME types and re-encoding raw image buffers via GD library to neutralize steganographic payloads.

SKILLS

- **Languages:** C, C++, Java, Python, JavaScript, HTML, PHP
- **Fundamentals:** Data Structures & Algorithms, OOPs, Operating Systems, DBMS
- **AI Frameworks & other Libraries:** PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Pandas, NumPy, Hugging Face Transformers, PEFT (LoRA)
- **Other technologies:** PostgreSQL, Apache, Linux, Wireshark

POSITIONS OF RESPONSIBILITY

- Senior Teaching Assistant for CS2233 Data Structures
- Teaching Assistant for ID1050 Intro to Artificial Intelligence
- Teaching Assistant for ID1063 Intro to Programming

RELEVANT COURSES

- **ML & Data Science**: Foundations of Machine Learning, Deep Learning, Natural Language Processing, Data Mining
- **Networks**: Advanced Computer Networks, Network Security
- **Other courses**: Advanced Data Structures & Algorithms, Linear Optimization

EXTRACURRICULAR

- Part of the Pokémon community @ IITH (GO & Showdown)